

Geometry

Unit 5

Lesson 8 Homework

Name Answer Key

Date _____

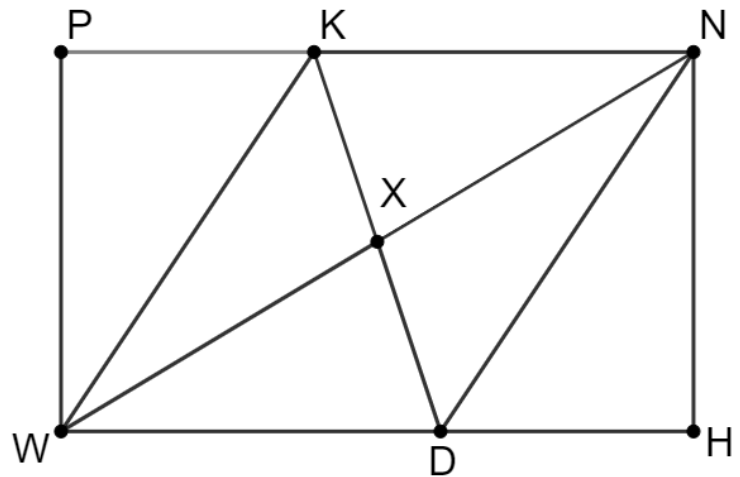
Period _____

Quadrilateral $PNHW$ is shown where $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$, and $\overline{PW} \cong \overline{NH}$.

Given: $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$, $\overline{PW} \cong \overline{NH}$

Prove: $\overline{PN} \cong \overline{WH}$

Use the given information to answer the questions about the incomplete proof that is shown.



Statement	Reason
1. $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$	1. Given
2. $\angle WPN$ and $\angle NHW$ are right angles	2.
3.	3. definition of right angles
4.	4.
5.	5. definition of congruence
6. $\overline{PW} \cong \overline{NH}$	6. Given
7.	7.
8.	8.
9. $\overline{PN} \cong \overline{WH}$	9.

1. What is the reason for line 2 of the proof?

definition of perpendicular lines

2. What is the statement for line 3 of the proof?

$m\angle WPN = 90^\circ$ and $m\angle NHW = 90^\circ$

3. What is the statement for line 4 of the proof?

$$m\angle WPN = m\angle NHW$$

4. What is the reason for line 4 of the proof?

transitive property of equality

5. What is the statement for line 5 of the proof?

$$\angle WPN \cong \angle NHW$$

6. What is the statement for line 7 of the proof?

$$\overline{NW} \cong \overline{NW}$$

7. What is the reason for line 7 of the proof?

reflexive property of congruence

8. What is the statement for line 8 of the proof?

$$\triangle WPN \cong \triangle NHW$$

9. What is the reason for line 8 of the proof?

$$HL \cong$$

10. What is the reason for line 9 of the proof?

Corresponding Parts of Congruent Triangles are Congruent